



Partial Picking System

Quality Assurance Test Results

Comprehensive Test Results for Production Deployment Review

Version 1.3.0 | January 16, 2026

Prepared By: Wind - Developer

1. Executive Summary

Test Category	Total	Passed	Failed	Rate
Unit Tests	74	74	0	100%
Load Tests (30 Workstations)	30	30	0	100%
Concurrent Read Tests	50	50	0	100%
Weight Validation Test	1	1	0	100%
TOTAL	155	155	0	100%

Performance Metrics

Metric	Target	Actual	Status
P50 Latency	< 200ms	113 ms	Met
P95 Latency	< 500ms	155 ms	Met
Concurrent Workstations	30+	30	Met
Success Rate	> 80%	100%	Exceeded

2. Unit Tests - FEFO Selection (6 tests)

Goal: Ensure the system always picks lots with the oldest expiry date first (First Expired First Out).

Test	Parameter Tested	Expected Result
test_fefo_sort_by_expiry_date_ascending	Multiple lots with different expiry dates	Oldest expiry lot selected first
test_filter_partial_pick_from_multiple_lots	Need 10kg, Lot A=5kg, Lot B=5kg	Can pick 5kg from each lot
test_filter_includes_only_tfc1_location	Lots from TFC1 location	Only TFC1 lots included
test_filter_excludes_zero_available_quantity	Lot with 0 available qty	Lot excluded from selection
test_filter_includes_only_allowed_lot_status	Lots with various status codes	Only allowed status included
test_complete_fefo_workflow_with_production_data	Real production data	Complete workflow succeeds

Result: cargo test --test fefo_tests → 6 passed, 0 failed

3. Unit Tests - Weight Validation (12 tests)

Goal: Ensure picked weight is within tolerance range (from INMAST.User9). Reject over/under weight picks.

Test	Parameter Tested	Expected Result
test_weight_exactly_at_target_accepted	Weight = Target exactly	Pick accepted
test_weight_within_upper_tolerance_accepted	Weight = Target + tolerance	Pick accepted
test_weight_within_lower_tolerance_accepted	Weight = Target - tolerance	Pick accepted
test_weight_above_tolerance_rejected	Weight > Target + tolerance	Pick REJECTED
test_weight_below_tolerance_rejected	Weight < Target - tolerance	Pick REJECTED
test_tolerance_from_inmast_user9	Tolerance ±0.025 from INMAST.User9	Exact value used, not %
test_zero_tolerance_requires_exact_match	Tolerance = 0%	Only exact weight accepted
test_large_tolerance_allows_wider_range	Tolerance = 10%	Wider range accepted
test_precision_handling_three_decimals	Weight with 3 decimal places	Precision maintained
test_negative_weight_rejected	Negative weight value	Pick REJECTED
test_production_scenario_insalt02_typical_variance	Real INSALT02 item data	Variance handled correctly
test_constitutional_compliance_no_override_allowed	Attempt to bypass validation	Override NOT allowed

Result: cargo test --test validation_tests → 12 passed, 0 failed

4. Unit Tests - Atomic Transaction (11 tests)

Goal: Ensure all 6 database phases execute atomically. If any phase fails, ALL phases rollback.

Test	Parameter Tested	Expected Result
test_all_phases_execute_successfully	Valid pick request	All 6 phases complete
test_rollback_on_phase1_failure	Phase 1 INSERT fails	All phases rolled back
test_rollback_on_phase2_failure	Phase 2 UPDATE fails	All phases rolled back
test_rollback_on_phase3_failure	Phase 3 INSERT fails	All phases rolled back
test_rollback_on_phase4_failure	Phase 4 UPDATE fails	All phases rolled back
test_audit_trail_preserved_on_rollback	Transaction rolled back	Audit fields NOT changed
test_phase_execution_order_enforced	Attempt out-of-order execution	Order strictly enforced
test_cannot_commit_with_failed_phase	One phase failed	Commit NOT allowed
test_production_scenario_successful_pick	Real run/item/lot data	Pick succeeds
test_production_scenario_duplicate_key_violation	Duplicate composite key	Error handled gracefully
test_constitutional_atomicity_guarantee	All phases	Either ALL or NONE complete

Result: cargo test --test transaction_tests → 11 passed, 0 failed

5. Unit Tests - Sequence Generation (14 tests)

Goal: Ensure unique LotTranNo sequence numbers are generated without duplicates or gaps.

Test	Parameter Tested	Expected Result
test_sequence_generation_returns_incremented_value	Request new sequence	Returns next value
test_sequence_generation_atomic_no_gaps	Multiple requests	No gaps in sequence
test_sequence_independent_types	Different sequence types	Each type independent
test_sequence_invalid_type_error	Invalid sequence type	Error returned
test_no_duplicates_sequential	Sequential requests	No duplicate numbers
test_no_duplicates_simulated_concurrent	100 concurrent requests	No duplicates
test_pallet_id_format_single_digit	Sequence = 1	Format: PAL-0000001
test_pallet_id_format_multiple_digits	Sequence = 123	Format: PAL-0000123
test_pallet_id_format_large_number	Sequence = 9999999	Format: PAL-9999999
test_pallet_id_format_overflow	Sequence > max	Overflow handled
test_bin_transfer_id_format	BT sequence request	Format: BT-XXXXXX
test_wrong_implementation_returns_old_value	Wrong SQL pattern	Detected and reported
test_production_sequence_workflow	Real workflow	Sequence works correctly
test_sequence_after_server_restart	Server restart	Previous sequence persists

Result: cargo test --test sequence_tests → 14 passed, 0 failed

6. Unit Tests - Audit Trail Preservation (12 tests)

Goal: Ensure audit fields (ItemBatchStatus, PickingDate, ModifiedBy) are NEVER overwritten during unpick.

Test	Parameter Tested	Expected Result
test_unpick_preserves_item_batch_status	Unpick operation	ItemBatchStatus preserved
test_unpick_item_batch_status_never_null	Set to NULL attempt	Field NOT set to NULL
test_unpick_preserves_picking_date	Unpick operation	PickingDate preserved
test_unpick_picking_date_never_null	Set to NULL attempt	Field NOT set to NULL
test_unpick_preserves_modified_by	Unpick operation	ModifiedBy preserved
test_unpick_modified_by_never_null	Set to NULL attempt	Field NOT set to NULL
test_unpick_resets_qty_to_zero	Unpick operation	PickedPartialQty = 0
test_unpick_clears_lot_no	Unpick operation	LotNo cleared
test_unpick_clears_bin_no	Unpick operation	BinNo cleared
test_wrong_unpick_violates_constitution	Wrong unpick SQL	Violation detected
test_production_unpick_scenario	Real unpick workflow	Audit preserved
test_multiple_unpick_preserves_original_audit_trail	Pick → Unpick → Pick	Original audit kept

Result: cargo test --test audit_trail_tests → 12 passed, 0 failed

7. Unit Tests - Composite Key Validation (19 tests)

Goal: Ensure composite primary keys (RunNo+RowNum+LineId) and (ItemKey+Location+LotNo) are validated.

Test	Parameter Tested	Expected Result
test_valid_composite_key_all_components	All key parts provided	Key valid
test_composite_key_missing_run_no	RunNo missing	Validation error
test_composite_key_missing_row_num	RowNum missing	Validation error
test_composite_key_missing_line_id	LineId missing	Validation error
test_lot_master_key_valid	ItemKey+Location+LotNo	Key valid
test_lot_master_key_missing_item_key	ItemKey missing	Validation error
test_lot_master_key_missing_location	Location missing	Validation error
test_lot_master_key_missing_lot_no	LotNo missing	Validation error
test_lot_master_key_empty_item_key	ItemKey = empty string	Validation error
test_partial_lot_picked_key_valid	All allocation key parts	Key valid
test_partial_lot_picked_key_missing_lot_no	LotNo missing	Validation error
test_partial_lot_picked_key_missing_bin_no	BinNo missing	Validation error
test_composite_key_zero_run_no_invalid	RunNo = 0	Validation error
test_composite_key_negative_row_num_invalid	RowNum < 0	Validation error
test_composite_key_zero_line_id_invalid	LineId = 0	Validation error
test_production_scenario_run_213972_batch_2	Run 213972 real data	Key works correctly
test_production_scenario_run_6000037	Run 6000037 real data	Key works correctly
test_production_lot_selection	Real lot selection	All keys validated
test_production_lot_allocation	Real lot allocation	All keys validated

Result: cargo test --test composite_key_tests → 19 passed, 0 failed

8. Load Tests (30 Workstations)

8.1 Test Configuration

Parameter	Value
Concurrent Workstations	30
Production Runs	6000003, 6000074, 6000086
Total Items	30 different items
Bin Restriction	PW* prefix only
Request Stagger	200ms per workstation
Weight Limit	50% of MaxWeight

8.2 6-Phase Atomic Transaction

Phase	Table	Operation	Purpose
1	Cust_PartialLotPicked	INSERT	Create lot allocation record
2	cust_PartialPicked	UPDATE	Update PickedPartialQty
3	LotTransaction	INSERT	Create audit trail
4	LotMaster	UPDATE	Update QtyCommitSales
5	PNITEM	UPDATE	Update ActualQty, Status
6	PNMAST	UPDATE	Update Status R to A

8.3 Test Source Code

```
fn get_30_workstation_test_data() -> Vec<(i32, i32, i32, &str, f64, &str, &str)> {
  vec![
    (6000003, 1, 2, "SPTURM01", 15.0, "2602538", "PWBF-06"),
    (6000074, 1, 1, "PXSUGB01", 19.337, "888845", "PWBE-13"),
    (6000086, 1, 3, "INBATR25", 21.2, "2601751", "PWBB-03"),
    // ... 30 items total across 3 runs
  ]
}
```

8.4 Results by Run

Run	Items	Successful	Failed	Rate
6000003	5	5	0	100%
6000074	19	19	0	100%
6000086	6	6	0	100%
TOTAL	30	30	0	100%

8.5 Latency Metrics

Metric	Value	Description
P50	113 ms	50% of picks within 113ms
P95	155 ms	95% of picks within 155ms
Duration	5.96 sec	All 30 concurrent picks

9. Constitutional Compliance

#	Requirement	Test Coverage	Status
1	Atomic 6-phase transaction	Transaction Tests (11)	Passed
2	FEFO lot selection	FEFO Tests (6)	Passed
3	Weight tolerance validation	Validation Tests (12)	Passed
4	Composite key uniqueness	Composite Key Tests (19)	Passed
5	Sequence number generation	Sequence Tests (14)	Passed
6	Audit trail preservation	Audit Trail Tests (12)	Passed
7	Concurrent workstation support	Load Tests (30)	Passed
8	No validation bypass	Validation Tests (12)	Passed

10. Summary

Category	Result
Total Tests	156
Passed	156
Failed	0
Pass Rate	100%
P50 Latency	113 ms
P95 Latency	155 ms
Workstations	30

Report: January 16, 2026 | Environment: Docker v1.3.0 | Database: SQL Server 2017 | By: Wind